Data Structures

CSL202

Quiz Set-A

1. Given the head of a singly linked list, group all the nodes with odd indices together followed by the nodes with even indices, and return the reordered list.

   The first node is considered odd, and the second node is even, and so on.

   Note that the relative order inside both the even and odd groups should remain as it was in the input.

   Input: [1,2,3,4,5]

   Output: [1,3,5,2,4]

2. Given the head of a sorted linked list, delete all duplicates such that each element appears only once. Return the linked list sorted as well. Input: [1,1,2,3,3]

   Output: [1,2,3]

3. Given the head of a singly linked list, return the middle node of the linked list.

   If there are two middle nodes, return the second middle node.

   Input: [1,2,3,4,5]

   Output: [3]

Data Structures

CSL202

Quiz Set-B

1. Given the head of a singly linked list and an integer k, split the linked list into k consecutive linked list parts.

   The length of each part should be as equal as possible: no two parts should have a size differing by more than one. This may lead to some parts being null.

   The parts should be in the order of occurrence in the input list, and parts occurring earlier should always have a size greater than or equal to parts occurring later.

   Return an array of the k parts.

   Input: [1,2,3,4,5,6,7,8,9,10], k = 3

   Output: [[1,2,3,4],[5,6,7],[8,9,10]]

2. Given the head of a linked list and an integer val, remove all the nodes of the linked list that has Node.val == val, and return the new head.

   Input: [1,2,6,3,4,5,6], val = 6

   Output: [1,2,3,4,5]

4. In a linked list of size n, where n is even, the ith node (0-indexed) of the linked list is known as the twin of the (n-1-i)th node, if 0 <= i <= (n / 2) - 1.

   For example, if n = 4, then node 0 is the twin of node 3, and node 1 is the twin of node 2. These are the only nodes with twins for n = 4.

   The twin sum is defined as the sum of a node and its twin.

   Given the head of a linked list with even length, return the maximum twin sum of the linked list.

   Input: [4,2,2,3]

   Output: 7

3. You are given the head of a linked list with n nodes.

   For each node in the list, find the value of the next greater node. That is, for each node, find the value of the first node that is next to it and has a strictly larger value than it.

   Return an integer array answer where answer[i] is the value of the next greater node of the ith node (1-indexed).

   If the ith node does not have a next greater node, set answer[i] = 0.

4. Given the pointer to the head node of a doubly linked list, reverse the order of the nodes in place. That is, change the next and prev pointers of the nodes so that the direction of the list is reversed.

   The initial doubly linked list is: 1 <-> 2 <-> 3 <-> 4

   The reversed doubly linked list is: 4 <-> 3 <-> 2 <-> 1

Data Structures

CSL202

Quiz Set-C

1. Given the head of a singly linked list, group all the nodes with odd indices together followed by the nodes with even indices, and return the reordered list.

   The first node is considered odd, and the second node is even, and so on.

   Note that the relative order inside both the even and odd groups should remain as it

Data Structures

CSL202

Quiz Set-D

1. In a linked list of size n, where n is even, the ith node (0-indexed) of the linked list is known as the twin of the (n-1-i)th node, if 0 <= i <= (n / 2) - 1.

   For example, if n = 4, then node 0 is the twin of node 3, and node 1 is the twin of node 2. These are the only nodes with twins for n = 4.

was in the input.

Input: [1,2,3,4,5]

Output: [1,3,5,2,4]

2. Given the head of a linked list, rotate the list to the right by k places.

Input: [1,2,3,4,5], k = 2

Output: [4,5,1,2,3]

3. Given the head of a linked list and an integer val, remove all the nodes of the linked list that has Node.val == val, and return the new head.

Input: [1,2,6,3,4,5,6], val = 6

Output: [1,2,3,4,5]

4. You are given the heads of two sorted linked lists list1 and list2.

Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists.

Return the head of the merged linked list.

Input: [1,2,4], list2 = [1,3,4]

Output: [1,1,2,3,4,4]

The twin sum is defined as the sum of a node and its twin.

Given the head of a linked list with even length, return the maximum twin sum of the linked list.

Input: [4,2,2,3]

Output: 7

2. Given the head of a sorted linked list, delete all duplicates such that each element appears only once. Return the linked list sorted as well.

Input: [1,1,2,3,3]

Output: [1,2,3]

3. Given the head of a linked list, rotate the list to the right by k places.

Input: [1,2,3,4,5], k = 2

Output: [4,5,1,2,3]

4. You are given the head of a linked list with n nodes.

For each node in the list, find the value of the next greater node. That is, for each node, find the value of the first node that is next to it and has a strictly larger value than it.

Return an integer array answer where answer[i] is the value of the next greater node of the ith node (1-indexed).

If the ith node does not have a next greater node, set answer[i] = 0.